0.0.1 Polynomial Ring Theory

Definition 0.0.1. Let R be a Commutative Ring with unity. Define *Polynomial Ring*:

$$R[x] \overset{\text{def}}{=} \left\{ \sum_{i=0}^{n} a_i x^i \;\middle|\; n \geq 0, \;\; a_i \in R \right\}$$

Addition is defined by pointwise, and Multiplcation is defined by:

$$\left(\sum_{i=0}^{n} a_i x^i \right) \times \left(\sum_{i=0}^{m} b_i x^i \right) = \sum_{k=0}^{n+m} \left(\sum_{i=0}^{k} a_i b_{k-i} \right) x^k$$

Proposition 0.0.2. Let R be an integral domain, and $p, q \in R[x]$ be non-zero elements.

1. $\deg(pq) = \deg p + \deg q$.
2. $R[x]$ is an integral domain.
3. If $p \in R[x]$ is unit, then $\deg p = 0$ and p is unit in R.

Proof.

$$\left(\sum_{i=0}^{n} a_i x^i \right) \left(\sum_{i=0}^{m} b_i x^i \right) = a_n b_m x^{n+m} + \cdots ,$$

This proves the statement (1) immediately, and assume not zero, then proves $2)$.
And, if $p \in R[x]$ is unit, then

$$0 = \deg 1 = \deg(pp^{-1}) = \deg p + \deg p^{-1}$$

Thus, $\deg p = \deg p^{-1} = 0$ and this implies $p, p^{-1} \in R$. □

0.0.2 Basic Theorems

> **Theorem 0.0.3.** Let I be an ideal of the Commutative Ring R with unity, and $(I) \subseteq R[x]$.
> Then, $R[x]/(I) \cong (R/I)[x]$. In particular, if I is prime ideal in R, then (I) is prime ideal in $R[x]$.

Proof. First, establish that $(I) = I[x]$. Since properties of Ideal, $(I) = IR[x] = I[x]$ directly.

Now, define a map $\varphi : R[x] \to (R/I)[x] : \sum_{i=0}^{n} a_i x^i \mapsto \sum_{i=0}^{n}(a_i + I)x^i$. Then, φ is homomorphism with $\ker \varphi = I[x]$.

The first-iso. Thm gives $R[x]/(I) = R[x]/I[x] \cong (R/I)[x]$. Particular,

$$I \text{ prime ideal} \iff R/I \text{ integral domain} \implies (R/I)[x] = R[x]/(I) \text{ integral domain} \iff (I) \text{ prime ideal.}$$

□

> **Theorem 0.0.4.** If F is a field, then $F[x]$ is Euclidean domain.
> Specifically, assume R is Commutative Ring with unity, $f, g \in R[x]$ with $\deg f, \deg g \geq 0$.
> If leading coefficient of g is unit in R, then there exists unique $q, r \in R[x]$ such that
>
> $$f(x) = g(x)q(x) + r(x) \quad (\deg r(x) < \deg g(x))$$

Proof. If $\deg f < \deg g$, put $g(x) = 0$ and $r(x) = f(x)$. Then proved. Suppose that $\deg f \geq \deg g$, and using induction.
If $\deg f = 0$, then put $g(x) = 0$, write leading coefficient of g as b and of f as a. Then, put $q = b^{-1}a$, $r = 0$.
If $\deg f \geq 1$, put $n = \deg f$, $m = \deg g$. Then $n \geq m$. Write:

$$\begin{aligned} f(x) &= a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \\ g(x) &= b_m x^m + b_{m-1}x^{m-1} + \cdots + b_1 x + b_0 \end{aligned}$$

Then, by induction,

$$\begin{aligned} f(x) &= a_n b_m^{-1} x^{n-m} g(x) + f_1(x) && (\deg f_1 < n-1) \\ &= a_n b_m^{-1} x^{n-m} g(x) + q_1(x)g(x) + r(x) && (\deg r < \deg g) \\ &= (a_n b_m^{-1} x^{n-m} q_1(x))g(x) + r(x) && (\deg r < \deg g) \end{aligned}$$

To show the uniquness,

$$f = q_1 g + r_1 = q_2 g + r_2 \implies g(q_1 - q_2) = r_2 - r_1 \implies \deg g(q_1 - q_2) = \deg g + \deg(q_1 - q_2) = \deg(r_2 - r_1) < \deg g$$

□

0.0.3 Relations between Rings and Their Polynomial Rings

Lemma 0.0.5. *Gauss's Lemma*
Let R be a Unique Factorization Domain with field of fractions F.
If $p(x) \in R[x]$ is reducible in $F[x]$, then $p(x)$ is reducible in $R[x]$.

Proof. Let $p(x) \in R[x]$ be reducible in $F[x]$. i.e.,

$$p(x) = A(x)B(x) \text{ for some } A(x), B(x) \in F[x] \text{ with } A(x), B(x) \text{ are both non-zero and non-units.}$$

Both $\deg A$ and $\deg B$ are at least 1: if either degree were zero, then lie in F, hence a unit - contradiction.
Write $A(x) = \sum_{i=0}^{n} \frac{r_i}{a_i} x^i$ and $B(x) = \sum_{i=0}^{m} \frac{s_i}{b_i} x^i$., and put $d_1 = a_1 \cdots a_n$, $d_2 = b_1 \cdots b_m$.
Now, $d_1 d_2 p(x) = d_1 A(x) d_2 B(x)$ where $d_1 A(x), d_2 B(x) \in R[x]$.
If $d = d_1 d_2$ is unit in R, then $p(x) = (d^{-1} d_1 A(x))(d_2 B(x))$ where $d^{-1} d_1 A(x), d_2 B(x) \in R[x]$, both are non-unit.
Suppose that d is not unit. Write $d = p_1 p_2 \cdots p_n$ is factorization of d.
p_1 is prime, being irreducible in U.F.D. $(p_1) = p_1 R[x]$ is prime, $R[x]/p_1 R[x] \cong (R/p_1 R)[x]$ is an integral domain.
Since $dp(x) = p_1 \cdots p_n p(x) \in p_1 R[x]$,

$$\overline{0} = dp(x) + p_1 R[x] = d_1 A(x) d_2 B(x) + p_1 R[x] = \overline{d_1 A(x)} \times \overline{d_2 B(x)}$$

Since $p_1 R[x]$ is an integral domain, either $\overline{d_1 A(x)}$ or $\overline{d_2 B(x)}$ is zero. WLOG, let $\overline{d_1 A(x)} = d_1 A(x) + p_1 R[x] = \overline{0}$.
This means all coefficient of $d_1 A(x)$ lies in $p_1 R$. Thus, we can cancle p_1 in the equation $dp(x) = d_1 A(x) d_2 B(x)$.
In finite process, we obtain $p(x) = A'(x) B'(x)$ where $A'(x), B'(x) \in R[x]$ with

$$A'(x) = rA(x), \; B'(x) = sB(x) \text{ where } r, s \in F$$

□

Corollary 0.0.6. Let R be a Unique Factorization Domain with field of fractions F.
Suppose that the greatest common divisor of the coefficients of $p(x) \in R[x]$ is 1. Then,

$$p(x) \text{ is irreducible in } R[x] \textit{ if and only if } p(x) \text{ is irreducible in } F[x]$$

In particular, if $p(x)$ is an irreducible monic polynomial in $R[x]$, then it is also irreducible in $F[x]$.

Proof. By Contraposition of Gauss's Lemma, if $p(x)$ is irreducible in $R[x]$, then $p(x)$ is irreducible in $F[x]$.
Conversely, suppose that $p(x)$ is reducible in $R[x]$, and the greatest common divisor of coefficients of $p(x)$ is 1.
Write $p(x) = a(x)b(x)$ where neitehr $a(x)$ nor $b(x)$ are not unit in $R[x]$, being reducible.
And, both $a(x)$ and $b(x)$ are not constant: because g.c.d. is 1. Thus, both are not unit in $F[x]$. □

> **Theorem 0.0.7.** R is Unique Factorization Domain *if and only if* $R[x]$ is Unique Factorization Domain.

Proof. Suppose that R is Unique Factoization Domain with field of fractions F.
Let $p(x) \in R[x]$ be non-zero element, and $d \in R$ be the greatest common divisor of coefficients of $p(x)$.
Then, $p(x) = dp'(x)$ where g.c.d. of coefficient of $p'(x)$ is 1. More precisely, write $p(x) = \sum_{i=0}^{n} a_i x^i,\ (a_i \in R)$.

$$p(x) = \sum_{i=0}^{n} a_i x^i = \sum_{i=0}^{n} d a_i' x^i = d\left(\sum_{i=0}^{n} a_i' x^i\right)$$

for some $a_i' \in R$ such that $a_i = da_i'$. Put g.c.d of a_i''s to $d' \in R$. Then, $a_i = da_i' = dd'a_i''$.
This implies dd' is divides every a_i; hence dd' divides d. That is, d' is unit, thus d' must be 1.
Since $F[x]$ is U.F.D, let $p'(x) = p_1(x)p_2(x)\cdots p_n(x)$ be a factorization of $p(x)$ in $F[x]$.
The g.c.d of $p'(x)$ is 1, thus g.c.d. of each $p_i(x)$ is 1.
Now, the corollary of the Gauss's Lemma gives that every $p_i(x)$ is irreducible in $R[x]$.
Hence, $p'(x) = p_1(x)p_2(x)\cdots p_n(x)$ is irreducible factorization in $R[x]$. To show that uniqueness, let

$$p'(x) = p_1(x)\cdots p_n(x) = q_1(x)\cdots q_m(x)$$

are two irreducibles factorizations of $p'(x)$ in $R[x]$. Since g.c.d of $p'(x)$ is 1, each $p_i(x)$ and $q_j(x)$ have g.c.d. 1.
Since the corollary of the Gauss's Lemma, all factors are irreducibles in $F[x]$ and $F[x]$ is U.F.D, $n = m$.
Moreover, each $p_i(x)$ and $q_i(x)$ are associates in $F[x]$(index rearrangement). Since associates up to unit in $F[x]$,

$$p_i(x) = \frac{a}{b} q_i(x) \quad \text{for some } a, b \in R^{\times}$$

That is, $bp_i(x) = aq_i(x)$; g.c.d. of left polynomial is b, and g.c.d. of right polynomial is a.
In integral domain, g.c.d. is unique up to unit, $a = ub$ for some unit $u \in R^{\times}$. That is,

$$bp_i(x) = aq_i(x) = ubq_i(x) \implies p_i(x) = uq_i(x)$$

Proof complete.

□

> **Theorem 0.0.8.** Let R be a Commutative Ring with identity.
> If $R[x]$ is Principal Ideal Domain, then R is a Field.

Proof. (Conti.) □

0.0.4 Irreducibility Criteria

Lemma 0.0.9. Let F be a field, and $p(x) \in F[x]$.

$p(x)$ has a factor of degree one *if and only if* $p(x)$ has a root in F

Proof. Trivial. □

Lemma 0.0.10. Let F be a field, and $p(x) \in F[x]$ be a polynomial of degree 2 or 3.

$p(x)$ is reducible *if and only if* $p(x)$ has a root in F.

Proof. Trivial. □

Theorem 0.0.11. Let $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \in \mathbb{Z}[x]$.
If $\dfrac{r}{s} \in \mathbb{Q}$ is a root of $p(x)$ where r and s are relatively prime, then $r \mid a_0$ and $s \mid a_n$.
In particular, if $p(x) \in \mathbb{Z}[x]$ is *monic polynomial* and $p(d) \neq 0$ for all $d \mid p(0)$, then $p(x)$ has no root in $\mathbb{Q}$.

Proof. By hypothesis,

$$\begin{aligned} & p\left(\frac{r}{s}\right) = a_n \left(\frac{r}{s}\right)^n + a_{n-1}\left(\frac{r}{s}\right)^{n-1} + \cdots + a_1\left(\frac{r}{s}\right) + a_0 = 0 \\ \implies & a_n r^n + a_{n-1}r^{n-1}s + \cdots + a_1 r s^{n-1} + a_0 s^n = 0 \\ \implies & a_n r^n = -(a_{n-1}r^{n-1}s + \cdots + a_1 r s^{n-1} + a_0 s^n = s(a_{n-1}r^{n-1} + \cdots + a_1 r s^{n-2} + a_0 s^{n-1}) \end{aligned}$$

Hence, s divides $a_n r^n$ and s and r are relatively prime, $s \mid a_n$. Similarly, $r \mid a_0$.
And, the second assertion is clear by contraposition. □

Theorem 0.0.12. Let $I \subsetneq R$ be a proper ideal of integral domain R, and $p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \in R[x]$.
If the image of $p(x)$

$$\tilde{p}(x) = x^n + (a_{n-1} + I)x^{n-1} + \cdots + (a_1 + I)x + (a_0 + I) \in (R/I)[x]$$

cannot be factored two polynomials into two smaller degree,
then $p(x)$ is irreducible in $R[x]$.

Proof. Suppose that $p(x)$ is reducible in $R[x]$.
Then, there exist two monic $a(x), b(x) \in R[x]$ with $\deg a, \deg b \geq 1$ such that $p(x) = a(x)b(x)$.
But, $p(x) + I[x] = a(x)b(x) + I[x] = (a(x) + I[x]) \times (b(x) + I[x])$ is still reducible in $(R/I)[x]$, because:
leading coefficient of a_n is $1 \implies a(x) + I[x]$ is not constant, being unit 1 cannot be in the proper ideal I. □

Theorem 0.0.13. *Eisenstein's Criterion*
Let P be a prime ideal of the integral domain R, and $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \in R[x], \ (n \geq 1)$.
Suppose $a_{n-1}, \ldots, a_1, a_0$ are contained in P and a_0 is not contained in P^2. Then, $f(x)$ is irreducible in $R[x]$.

Proof. Proof by Contradiction. Suppose $f(x)$ is reducible. Then, $f(x) = a(x)b(x)$
for some nonconstant $a(x), b(x)$ in $R[x]$ (If it is constant, then contradicts to monic.)
Observe that: In $(R/P)[x]$,

$$(1 + P)x^n + (a_{n-1} + P)x^{n-1} + \cdots + (a_1 + P)x + (a_0 + P) = x^n = \overline{a(x)b(x)} \in (R/P)[x] \cong R[x]/P[x]$$

Thus, the constant terms of $a(x)$ and $b(x)$ both are contained in P.
But this implies that the product of these two constants is contained in P^2, contradiction. □

0.0.5 Summary and Diagram

$$
\begin{tikzcd}
R & \text{Field} \arrow[r, Rightarrow] \arrow[ddr, Rightarrow] & \text{Euclidean Domain} \arrow[r, Rightarrow] & \text{P.I.D.} \arrow[rr, Rightarrow] \arrow[dr, Rightarrow] & & \text{U.F.D.} \arrow[r, Rightarrow] \arrow[dd, Leftrightarrow] & \text{Integral Domain} \arrow[dd, Leftrightarrow] \\
 & & & & \text{Noetherian Domain} \arrow[dd, Leftrightarrow] & & \\
R[x] & & \text{Euclidean Domain} \arrow[r, Rightarrow] & \text{P.I.D.} \arrow[rr, Rightarrow] \arrow[dr, Rightarrow] \arrow[uull, Rightarrow] & & \text{U.F.D.} \arrow[r, Rightarrow] & \text{Integral Domain} \\
 & & & & \text{Noetherian Domain} & &
\end{tikzcd}
$$